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JMeter Performance Testing & CI/CD Integration

1) Create the JMeter Test Plan : Create a JMeter test plan containing five Thread Groups.

2) Configure the Thread Groups

Configure each Thread Group with:

- Number of Users: **10**
- Ramp-Up Period: **5 seconds**
- Loop Count: **3**

3) Configure Test Scenarios: Add the required HTTP Request samplers, test data, parameters, listeners and assertions for each application also

4) Execute the Test in Non-GUI Mode: Execute the JMeter test using the command line to perform the load test efficiently.

5) Analyze Results Using Aggregate Graph/Report

Open the generated JTL file in JMeter and analyze the performance results using the Aggregate Graph/Report.

6) Validate Performance Metrics

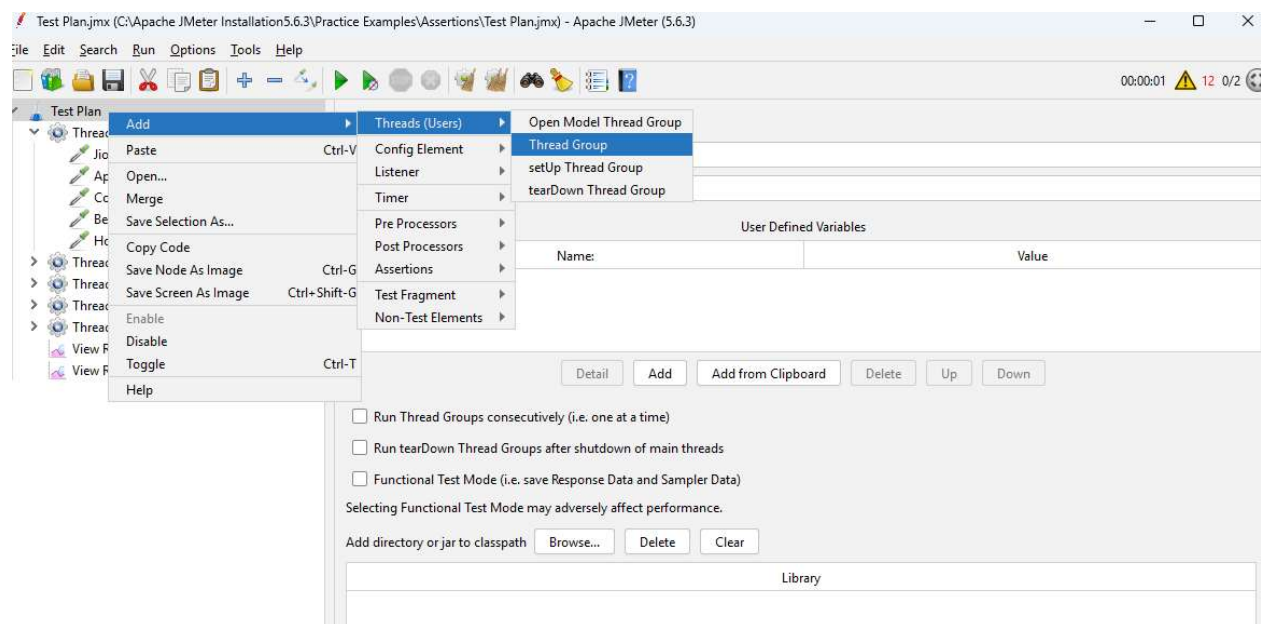
- Total Samples
- Throughput
- Latency
- Median
- Minimum Response Time
- Maximum Response Time
- 90th Percentile
- 95th Percentile
- Error Percentage

7) Generate the HTML Dashboard Report and Integrate with Jenkins

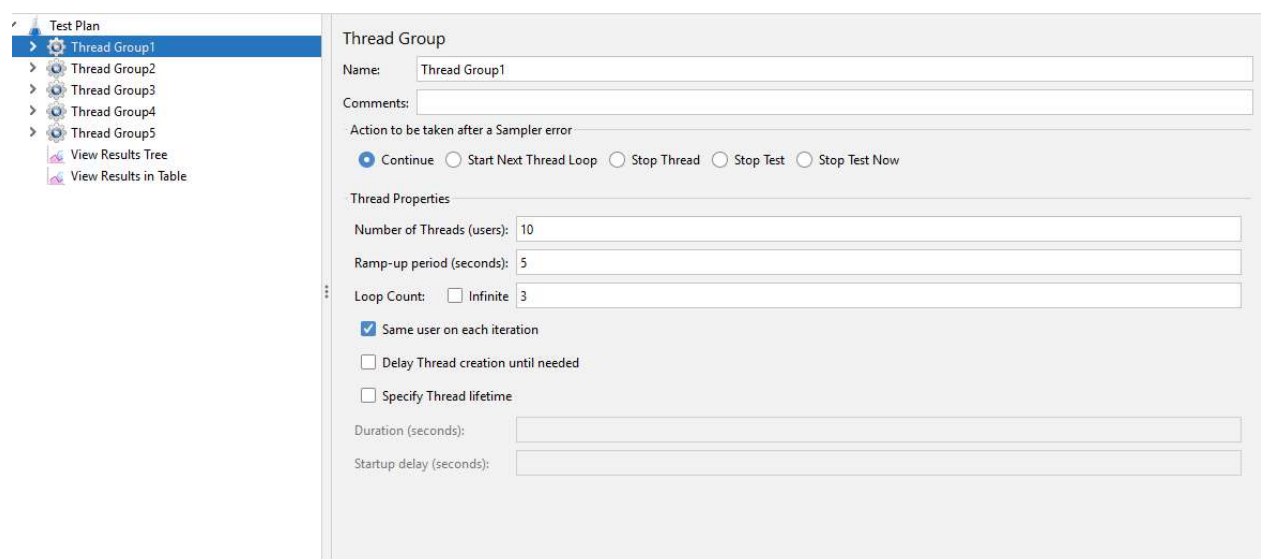
Use the JTL file to generate the JMeter **HTML Dashboard Report** containing graphical performance statistics.

8) Validate Jenkins Execution

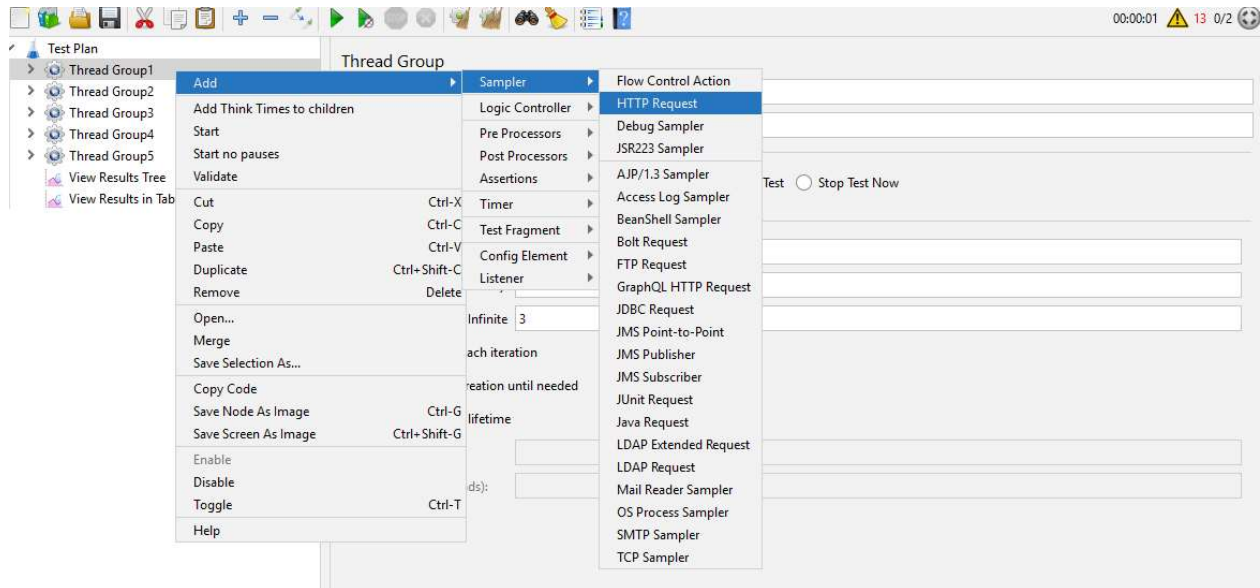
Step 1: Create a JMeter test plan containing five Thread Groups



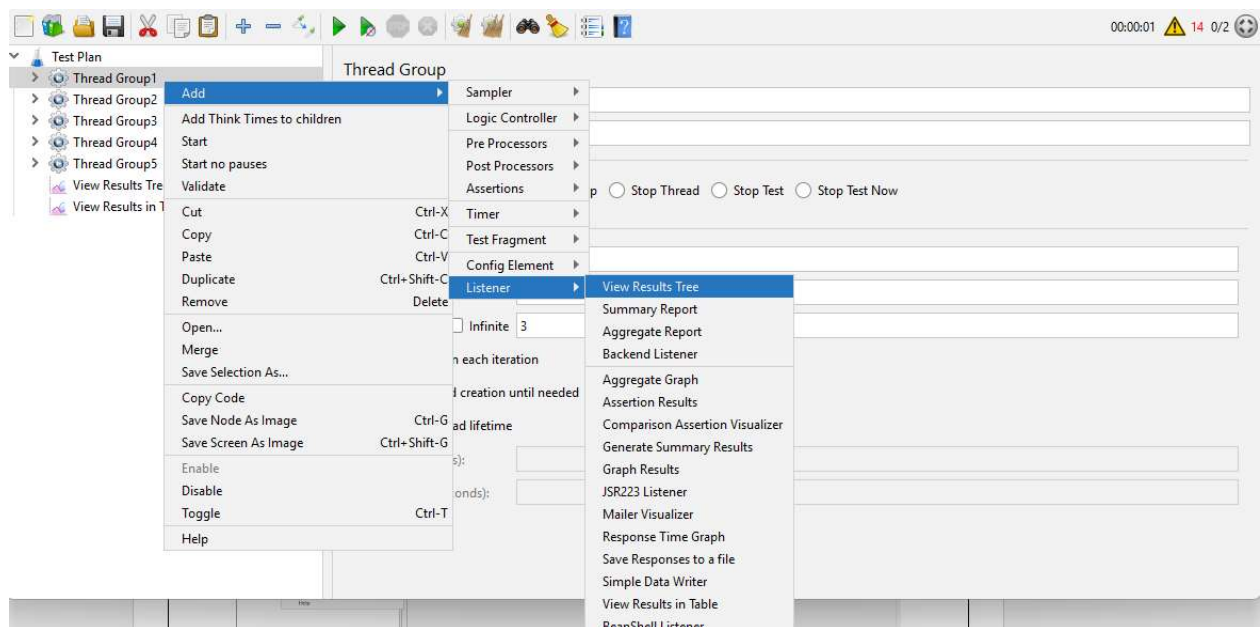
Step 2: Configure each Thread Group according to the requirements mentioned

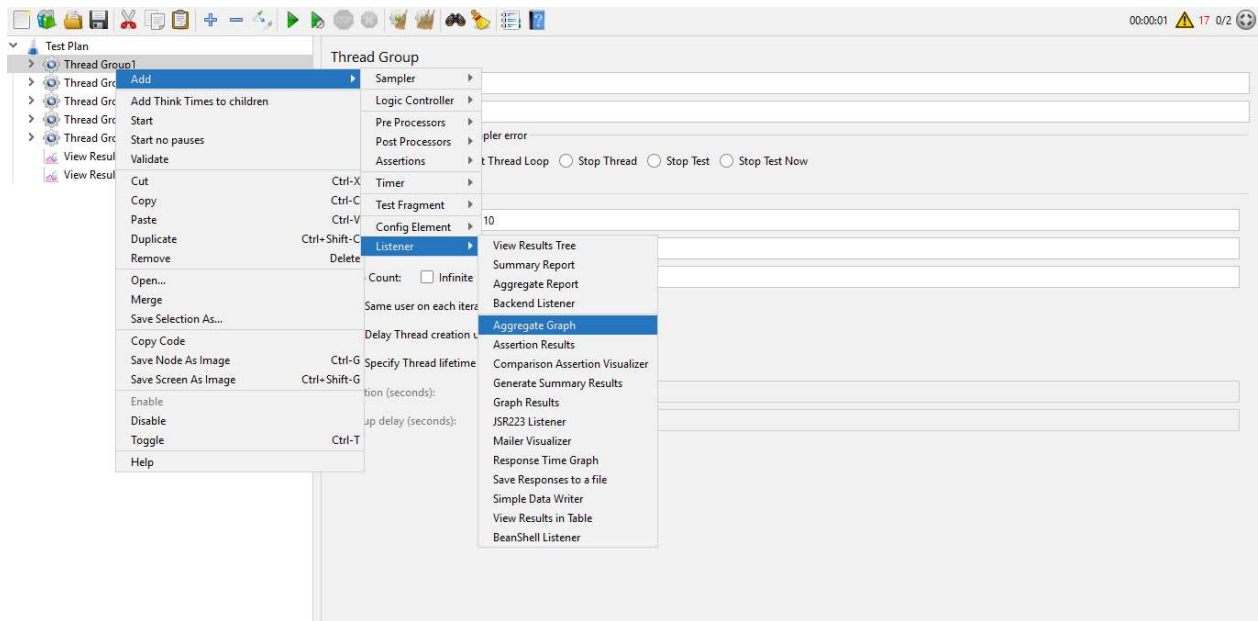
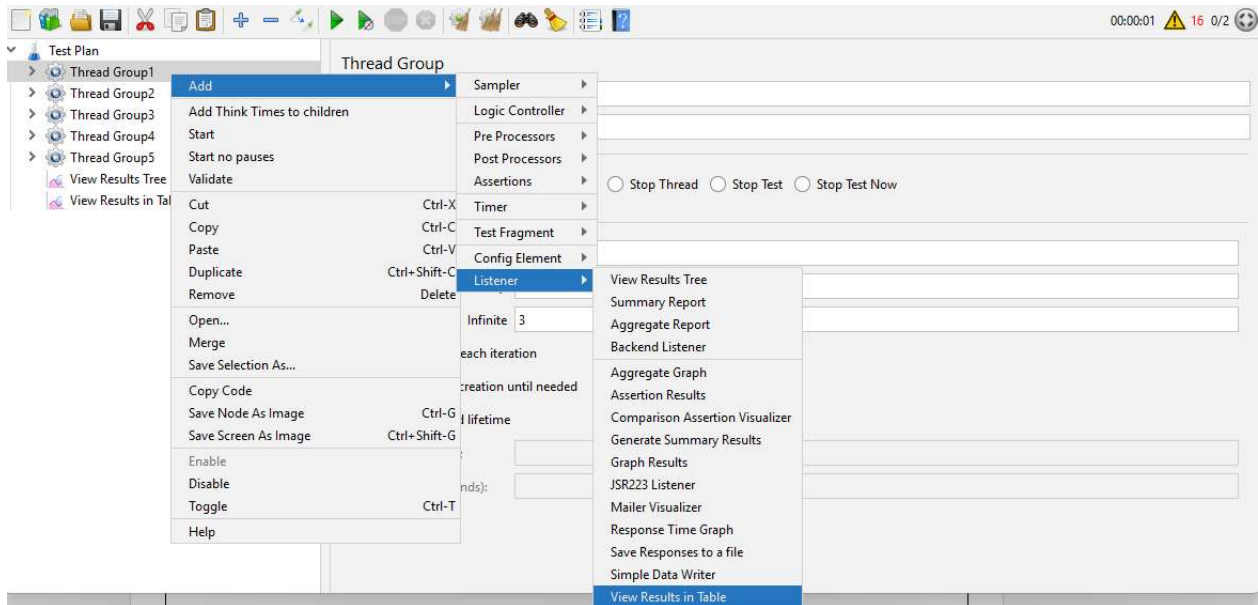


Step 3: Add samplers to each thread group using different E-commerce websites

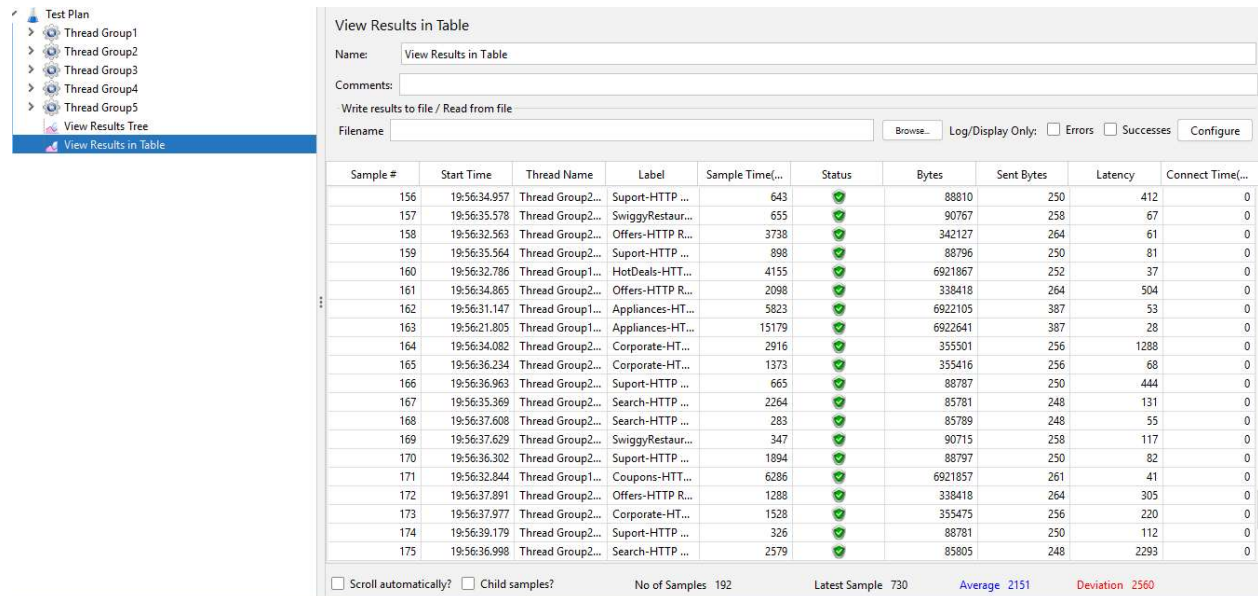


Step 4: Add listeners





Step 5 : Run and validate



View Results in Table

Name: View Results in Table

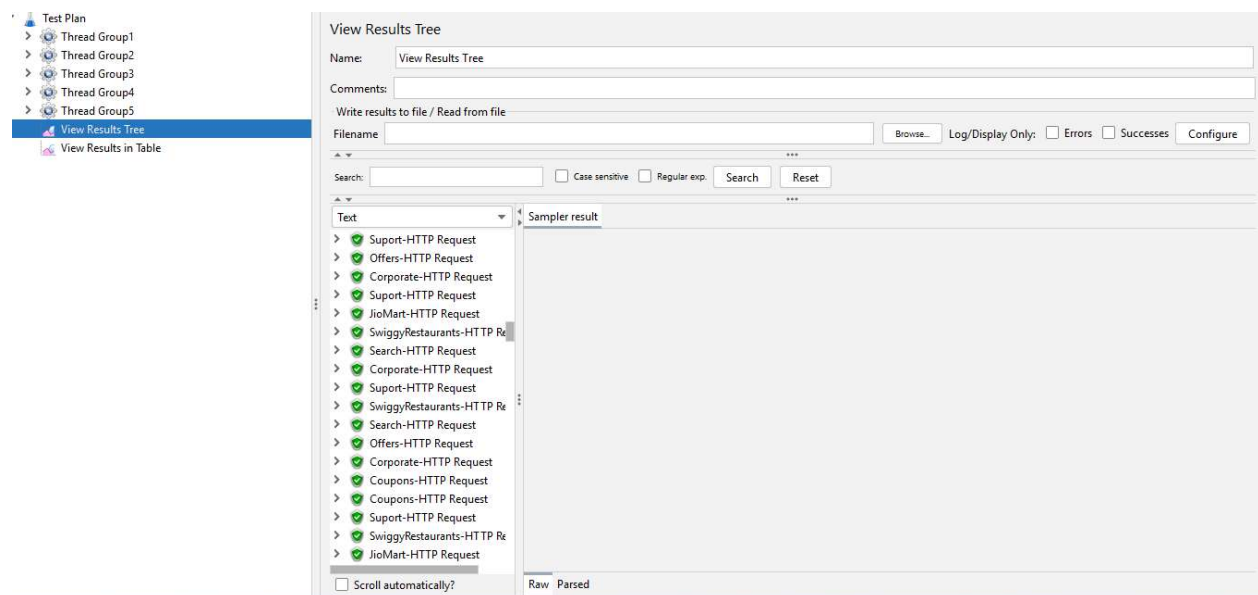
Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Sample #	Start Time	Thread Name	Label	Sample Time...	Status	Bytes	Sent Bytes	Latency	Connect Time...
156	19:56:34.957	Thread Group2...	Support-HTTP ...	643	✓	88810	250	412	0
157	19:56:35.578	Thread Group2...	SwiggyRestaur...	655	✓	90767	258	67	0
158	19:56:32.563	Thread Group2...	Offers-HTTP R...	3738	✓	342127	264	61	0
159	19:56:35.564	Thread Group2...	Support-HTTP ...	898	✓	88796	250	81	0
160	19:56:32.786	Thread Group1...	HotDeals-HTT...	4155	✓	6921867	252	37	0
161	19:56:34.865	Thread Group2...	Offers-HTTP R...	2098	✓	338418	264	504	0
162	19:56:31.147	Thread Group1...	Appliances-HT...	5823	✓	6922105	387	53	0
163	19:56:21.805	Thread Group1...	Appliances-HT...	15179	✓	6922641	387	28	0
164	19:56:34.082	Thread Group2...	Corporate-HT...	2916	✓	355501	256	1288	0
165	19:56:36.234	Thread Group2...	Corporate-HT...	1373	✓	355416	256	68	0
166	19:56:36.963	Thread Group2...	Support-HTTP ...	665	✓	88787	250	444	0
167	19:56:35.369	Thread Group2...	Search-HTTP ...	2264	✓	85781	248	131	0
168	19:56:37.608	Thread Group2...	Search-HTTP ...	283	✓	85789	248	55	0
169	19:56:37.629	Thread Group2...	SwiggyRestaur...	347	✓	90715	258	117	0
170	19:56:36.302	Thread Group2...	Support-HTTP ...	1894	✓	88797	250	82	0
171	19:56:32.844	Thread Group1...	Coupons-HTT...	6286	✓	6921857	261	41	0
172	19:56:37.891	Thread Group2...	Offers-HTTP R...	1288	✓	338418	264	305	0
173	19:56:37.977	Thread Group2...	Corporate-HT...	1528	✓	355475	256	220	0
174	19:56:39.179	Thread Group2...	Support-HTTP ...	326	✓	88781	250	112	0
175	19:56:36.998	Thread Group2...	Search-HTTP ...	2579	✓	85805	248	2293	0

☐ Scroll automatically? ☐ Child samples? No of Samples 192 Latest Sample 730 Average 2151 Deviation 2560



View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Search: Case sensitive Regular exp. Search Reset

Text

- Support-HTTP Request
- Offers-HTTP Request
- Corporate-HTTP Request
- Support-HTTP Request
- JioMart-HTTP Request
- SwiggyRestaurants-HTTP Request
- Search-HTTP Request
- Corporate-HTTP Request
- Support-HTTP Request
- SwiggyRestaurants-HTTP Request
- Search-HTTP Request
- Offers-HTTP Request
- Corporate-HTTP Request
- Coupons-HTTP Request
- Coupons-HTTP Request
- Support-HTTP Request
- SwiggyRestaurants-HTTP Request
- JioMart-HTTP Request

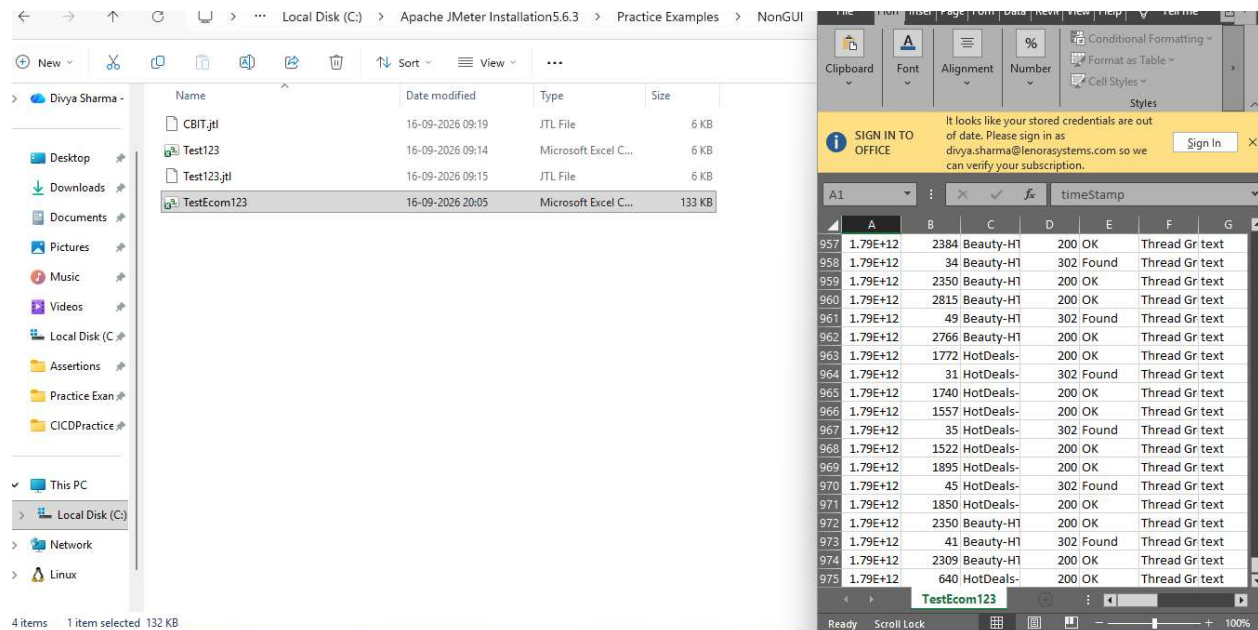
☐ Scroll automatically? Raw Parsed

Step 6:

1. Open Command Prompt.
2. Navigate to the JMeter bin folder using the following command:
3. `cd /d "C:\ApacheJmeter5.6.3\apache-jmeter-5.6.3\bin"`

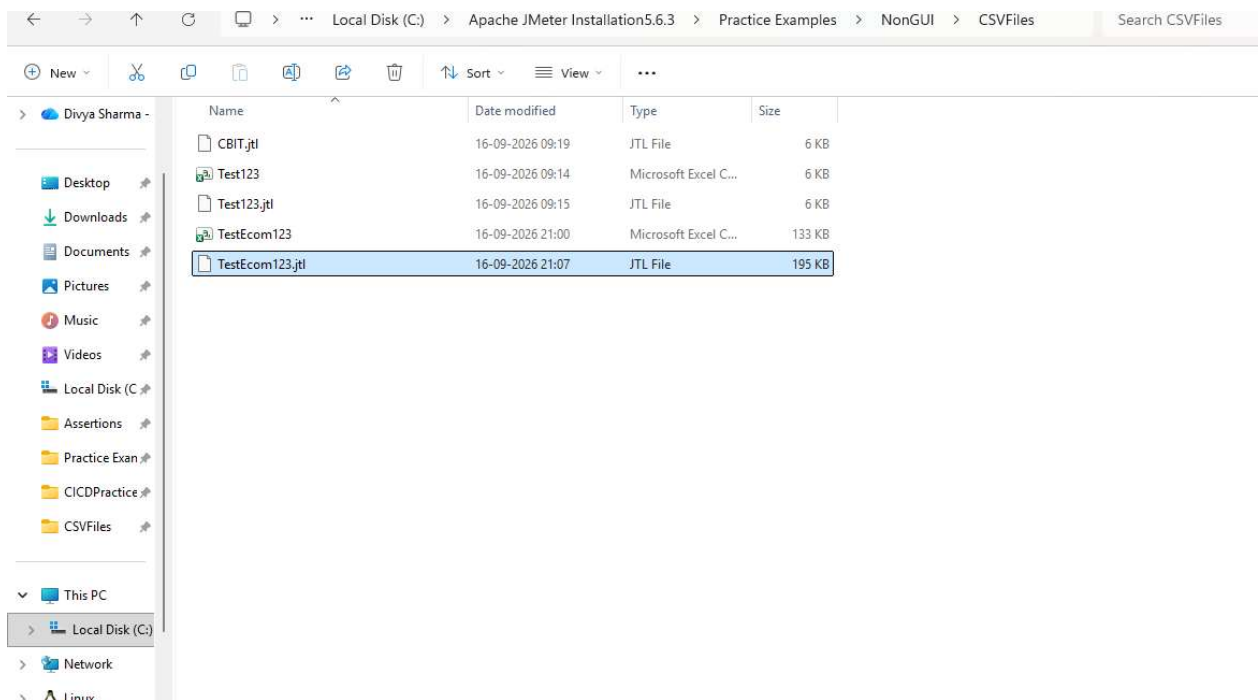
4. Generate the CSV result file by executing the JMeter test plan in non-GUI mode: `jmeter -n -t "C:\Apache JMeter Installation5.6.3\Practice Examples\NonGUI\Test Plan.jmx" -l "C:\Apache JMeter Installation5.6.3\Practice Examples\NonGUI\CSVFiles\TestEcom123.csv"`

The test execution results are saved in the specified location as a CSV file.

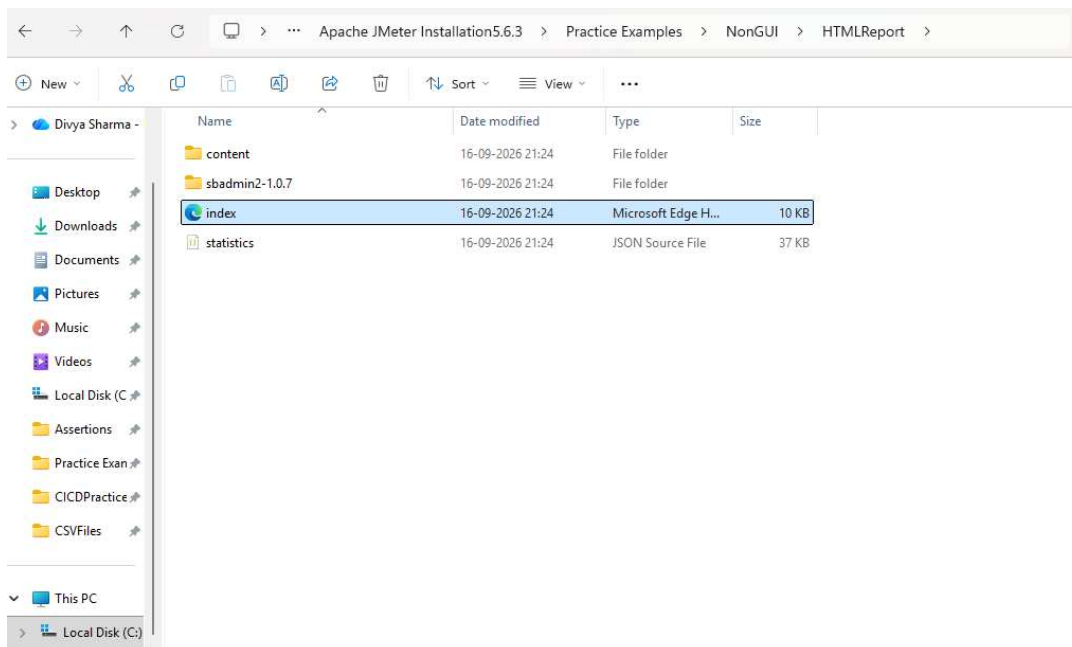


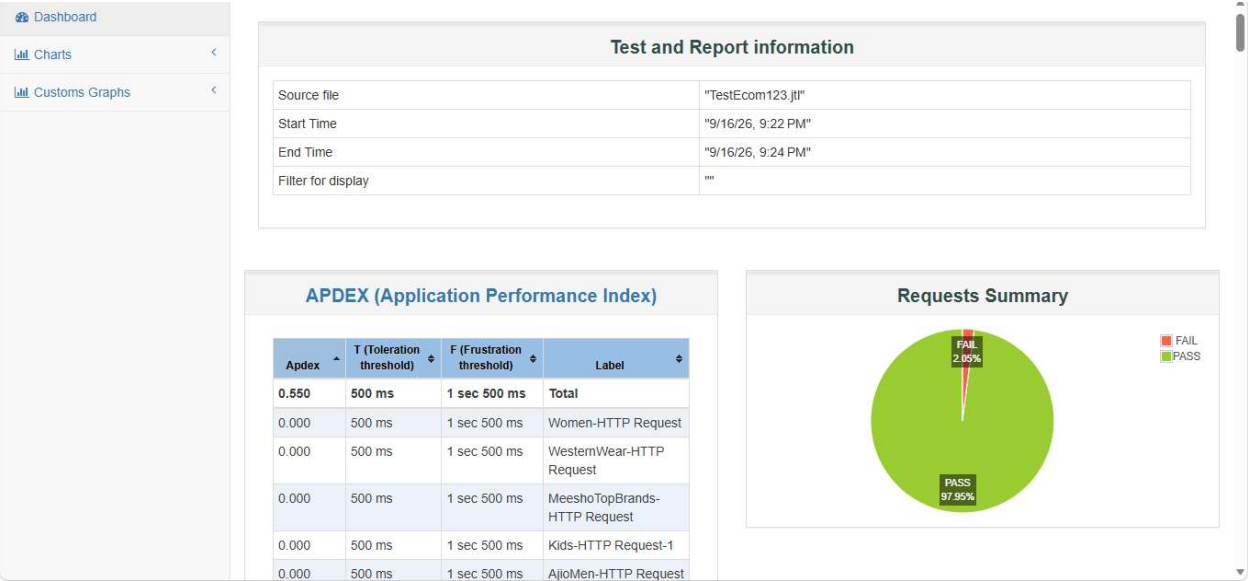
Step 7:

1. Generate the JTL file to use as the input for the Aggregate Graph: `jmeter -n -t "C:\JmeterPracticeExamples\NonGUI\Test Plan.jmx" -l "C:\JmeterPracticeExamples\NonGUI\CSVFiles\TestEcom123.csv"`
2. Open the generated JTL file in JMeter and use the Aggregate Graph/Report to analyze the performance results, including Throughput, Latency, Median, Minimum, Maximum, Percentiles, Total Samples, and Error Percentage.



Step 7: Html reports





Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	976	20	2.05%	2016.67	7	29322	521.50	5664.20	8460.80	14271.71	11.18	23896.23	1.91
AjioMen-HTTP Request	1	1	100.00%	785.00	785	785	785.00	785.00	785.00	785.00	1.27	1.28	0.31
AjioMen-HTTP Request-0	1	0	0.00%	662.00	662	662	662.00	662.00	662.00	662.00	1.51	0.20	0.18
AjioMen-HTTP Request-1	1	1	100.00%	119.00	119	119	119.00	119.00	119.00	119.00	8.40	7.34	1.05
Appliances-HTTP Request	30	0	0.00%	4814.93	762	13854	3632.00	10200.10	13022.95	13854.00	0.36	2433.15	0.14
Appliances-HTTP Request-0	30	0	0.00%	52.63	28	462	35.00	55.30	275.00	462.00	0.36	0.09	0.05

Step 8: Run through Jenkins – create freestyle project

New Item

Enter an item name

JenkinsCICDUsingJmeter1

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Multi-configuration project

OK

Step 9 : Configure Jenkins

Jenkins

JenkinsCICDUsingJmeter

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command

Command

See the list of available environment variables

cd "C:\Apache JMeter Installation5.6.3\apache-jmeter-5.6.3\bin"
jmeter -Jjmeter.save.saveservice.output_format=xml -n -t "C:\Apache JMeter Installation5.6.3\Practice Examples\CICDPracticeTests\Test Plan.jmx" -l
"C:\Users\Divya\.jenkins\workspace\JenkinsCICDUsingJmeter\CICDPracticeTests.jtl"

Advanced

+ Add build step

Save

Apply

Step 10: Analyze Performance Trend

Jenkins

JenkinsCICDUsingJmeter

Performance Trend

Status

Changes

Workspace

Build Now

Delete Project

Performance Trend

Configure

Rename

Builds

Filter

Today

#5 7:24 PM

Performance Trend

Last Report

Filter trend data

Test file: CICDPracticeTests.jtl

Response time

Percentage of errors

Trend report

Trend report: CICDPracticeTests.jtl

Build	Line 90.0(ms)	Line 95.0(ms)	Max(ms)	Median(ms)	Min(ms)	Percentage of errors	average	average kb	errors	total kb
#3	6885	8043	11112	2425	94	3	3096	3310.51	10	1042811.51
#4	7123	8598	58695	2425	94	3	3209	3310.49	20	2085609.59
#5	7112	8416	58695	2427	77	3	3201	3310.49	30	3128415.13

Step 11: JMeter Aggregate Graph – Performance Test Results

